07-BREADBOARDS-WIRES-BATTERIES-AND-POWER

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**TP4056 USB-C 5V 1A Lithium Battery Charging Module**	Lithium battery charging board with **dual protection IC** (DW01A + 8205A MOSFET); 18650 Li-ion / LiPo charger with overcharge, overdischarge & overcurrent protection; USB-C input	**8 Units**	**Consumable**
02	**ROBOWAY 4S 4A Polymer Lithium Ion Charger Type-C (Pack of 4)**	14.8V 18650 battery charging module; USB-C input with boost converter; LED charging status indicator; integrated overcharge and balanced protection for 4-series Li-ion battery packs	**2 Packs (8 Units)**	**Consumable**
03	**Electronic Spices LM2596 DC-DC Step Down Power Supply Module (3A, Pack of 2)**	High-efficiency 3A switching voltage regulator buck converter; wide input voltage 4.0V–40V DC; adjustable output 1.3V–37V DC (supports 24V, 12V, 5V, 3.3V/3V rails); high efficiency (>90%); multi-turn trimmer pot; powers ESP32, Arduino, servos & motors from batteries	**5 Packs (10 Units)**	**Consumable**
04	**Robocraze SD Card Module for Arduino SPI Interface (3.3V/5V Level Shifter)**	SPI Interface Micro SD / MicroSDHC Reader Writer Module with **onboard 3.3V/5V Level Shifter**; data logging and storage expansion board for ESP32, Arduino, and Raspberry Pi robotics & IoT projects	**5 Units**	**Consumable**
05	**Robocraze DS3231 RTC Module \	AT24C32 Real Time Clock Memory Board**	**ESP32 / Arduino / Raspberry Pi**	**5 Units**
06	**Half-Size Solderless Breadboard (400 Tie-Points)**	400 tie-point solderless breadboard with self-adhesive backing tape; dual power distribution buses	**30 Units**	**Consumable**
07	**Full-Size Solderless Breadboard (800 Tie-Points)**	830 tie-point solderless breadboard; dual center terminal strip with 4 power distribution rails; side interlocking tabs	**20 Units**	**Consumable**
08	**Self-Adhesive Mini Proto Shield Breadboard**	Compact 170-point mini breadboard designed for Arduino Proto-Shield mounting	**15 Units**	**Consumable**
09	**General-Purpose Perfboard (Zero-PCB) — Medium**	FR2 single-sided copper clad dotted prototype board; size 80× 55 mm; 2.54mm hole spacing	**30 Units**	**Consumable**
10	**General-Purpose Perfboard (Zero-PCB) — Large**	FR2 single-sided copper clad dotted prototype board; size 140× 90 mm; 2.54mm hole spacing	**20 Units**	**Consumable**
11	**Male Breakable Pin Headers (Berg Strips)**	40-pin single row male headers; standard 2.54mm pitch; gold/tin plated; snappable to any pin length	**30 Strips**	**Consumable**
12	**Female Pin Headers (Berg Strips)**	40-pin single row female socket headers; standard 2.54mm pitch; straight mount	**30 Strips**	**Consumable**
13	**Serplex 560pcs Jumper Wire Kit (14 Lengths, 2–125mm)**	High-quality pre-formed U-shape solderless breadboard jumper wire kit; 14 different standard wire lengths (2mm to 125mm); flexible insulated copper core in segmented plastic organizer box	**5 Kits**	**Consumable**
14	**Jumper Wires — Male to Male (M-M)**	20cm flexible multicolored 40-pin ribbon jumper wires with molded square male pins	**5 Packs (200 Wires)**	**Consumable**
15	**Jumper Wires — Male to Female (M-F)**	20cm flexible multicolored 40-pin ribbon jumper wires with male pin to female socket	**5 Packs (200 Wires)**	**Consumable**
16	**Jumper Wires — Female to Female (F-F)**	20cm flexible multicolored 40-pin ribbon jumper wires with female socket to female socket	**3 Packs (120 Wires)**	**Consumable**
17	**Alligator Clip Test Leads**	Double-ended insulated alligator clip jumper cables; 12-inch length; multicolored jackets	**3 Packs (30 Leads)**	**Consumable**
18	**Hook-Up Wire Spools**	Single-core insulated hook-up wire spools (100 metres Red + 100 metres Black per set)	**2 Sets (4 Spools)**	**Consumable**

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
19	**Electronic Spices 2 x 3.7V 18650 Battery Holder with Wire Leads**	Dual 18650 Li-ion battery holder case (series connection for 7.4V DC nominal output) with color-coded positive/negative wire leads	**15 Units**	**Consumable**
20	**Electronic Spices 18650 Battery Holder 3-Slot with Wire Leads**	3-slot 18650 lithium battery holder box (3.7V × 3 = 11.1V / 12V output) with heavy-duty wire leads for robotics and motor supply	**10 Units**	**Consumable**
21	**Electronic Spices Li-ion Battery Universal Charger (Dual 3.7V 18650)**	Universal dual-slot desktop battery charger for 3.7V 18650 cylindrical lithium rechargeable cells with automatic cut-off and LED status	**3 Units**	**Consumable**
22	**AC to DC 4-Slot 18650 Battery Charger with LED Indicator**	Heavy-duty 4-slot intelligent AC mains charger for 3.7V rechargeable Li-ion 18650 batteries; independent channel charging with LED indicators	**2 Units**	**Consumable**
23	**HD-8993A 8-Slot Battery Charger with LCD Display**	8-slot multi-chemistry intelligent battery charger with LCD status display; 6400mA max output across independent channels; supports 18650, 26650, AA, AAA, Ni-MH, and 3.7V Lithium-ion cylindrical rechargeable batteries	**1 Unit**	**Consumable**
24	**3.7V 1200mAh LiPo (Lithium Polymer) Rechargeable Battery**	3.7V 1S flat pouch rechargeable lithium polymer cell with built-in Protection Circuit Module (PCM) preventing overcharge/overdischarge; pre-soldered leads / 2-pin JST connector; ideal for wearable electronics, IoT sensors, Seeed Studio XIAO ESP32-S3 battery charging, and TP4056 single-cell charging circuits	**30 Pieces**	**Consumable**
25	**18650 3.7V Li-ion (Lithium-Ion) Rechargeable Battery Cell**	Standard industrial 18650 cylindrical rechargeable lithium-ion cells (18× 65 mm); high-capacity (2000-2600 mAh) flat-top / button-top; fits 2-slot (7.4V) and 3-slot (11.1V) battery holders, desktop 4-slot chargers, robotics chassis, and high-drain BO motors	**30 Pieces**	**Consumable**
26	**9V Battery Snaps with DC Barrel Plug**	Heavy-duty molded hard-plastic 9V battery connector clips terminating in a 5.5mm/2.1mm DC barrel jack	**20 Units**	**Consumable**
27	**USB Programming Cables — Type-A to Type-B**	High-shielded 1.2m USB 2.0 cable for Arduino Uno and Arduino Mega	**10 Cables**	**Consumable**
28	**USB Programming Cables — Type-A to Mini-B**	High-shielded 1.0m USB 2.0 cable for Arduino Nano and microcontroller boards	**10 Cables**	**Consumable**
29	**USB to DC Barrel Jack Converter Cables**	5V USB Type-A to 5.5mm × 2.1mm center-positive DC plug cable for running breadboards from power banks	**5 Cables**	**Consumable**
30	**High-Capacity Portable Power Banks**	10,000 mAh dual-port (5V 2.1A / 1A) portable rechargeable power bank with battery LED meter	**3 Units**	**Consumable**
31	**Regulated DC Wall Power Adapter — 5V 2A**	230V AC to 5V DC, 2000mA regulated switch-mode wall power adapter with 5.5mm DC barrel jack	**5 Units**	**Consumable**
32	**Regulated DC Wall Power Adapter — 12V 2A**	230V AC to 12V DC, 2000mA regulated switch-mode wall power adapter with 5.5mm DC barrel jack	**5 Units**	**Consumable**
33	**Workshop Power Extension Boards**	Heavy-duty 5-socket universal surge-protected power extension strip with master switch, indicator, and 5m cord	**3 Units**	**Consumable**
34	**Twisted-Pair Mains Distribution Wires**	Red-Black twisted flexible copper electrical distribution wire	**20 Metres**	**Consumable**
35	**AC/DC Lamp Bulb Holders**	Standard insulated screw/bayonet socket bulb holders for bench load experiments	**5 Units**	**Consumable**

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
36	**FEDUS 12V 1.5Amp DC Power Supply Adapter Converter**	12 Volt 1.5A SMPS regulated power adapter converter (100–240V AC input, 12V 1.5A DC output with standard 5.5mm DC barrel jack); for CCTV cameras, routers, DVR/NVR, LCD monitors, LED strips, motors and microcontrollers	**2 Units**	**Consumable**